# Nuclear Shell Model Hamiltonians on the Hyperspherical Lattice

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We present structurally sparse qubit Hamiltonians for the nuclear shell model, constructed by applying the GeoVac angular-sparsity framework to the harmonic oscillator (HO) single-particle basis with a Mayer–Jensen spin–orbit term. The HO shell closures $2, 8, 20, 40, 70, 112$ emerge directly from graph state counting, and the physical magic numbers $2, 8, 20, 28, 50, 82, 126$ are recovered with a spin–orbit strength $v_{ls}/\hbar\omega \approx 0.17$ and a Nilsson $\ell(\ell+1)$ correction $d_{\ell\ell}/\hbar\omega \approx 0.02$. We build the first two-species qubit encoding for the deuteron ($1p + 1n$, 16 qubits, 688 non-identity Pauli terms, 1-norm $\approx 384$ MeV) using the Minnesota nucleon–nucleon potential and Moshinsky–Talmi brackets, and extend the architecture to helium-4 ($2p + 2n$, 16 qubits, 828 Pauli terms) with intra-species antisymmetrization and proton–proton Coulomb repulsion. Despite a Hilbert-space dimension $12.25\times$ larger than the deuteron, He-4 requires only $1.20\times$ more Pauli terms — a direct consequence of the angular selection rules that carry through to multi-particle systems. A composed nuclear–electronic deuterium encoding (26 qubits) validates the multi-block architecture and quantifies the $\sim 10^{13}$ coefficient ratio between nuclear, electronic, and hyperfine scales. We state and apply the Fock projection rigidity theorem: the $S^3$ conformal projection that makes electronic Hamiltonians equivalent to a graph Laplacian is unique to the Coulomb potential and does *not* extend to HO or Woods–Saxon systems. For nuclear Hamiltonians, the GeoVac contribution is therefore the angular machinery and qubit encoding, not the conformal projection. All nuclear physics inputs (Minnesota parameters, spin–orbit strength, $\hbar\omega$) follow the standard shell-model literature; the quantitative binding energies at $N_{\text{shells}} = 2$ are far from experiment and should be read as encoding-validation benchmarks, not as a nuclear-structure calculation.

## I. INTRODUCTION

Nuclear quantum simulation is an active frontier in quantum computing [8–12]. Exact solvers (no-core shell model, NCSM [13, 14]) remain the gold standard for light nuclei $A \lesssim 16$, but scale factorially with the number of active nucleons, and their extension to medium-mass nuclei requires importance truncation or symmetry-adapted bases. On current and near-term quantum hardware, the practical bottleneck is the resource cost of the qubit Hamiltonian: the nuclear interaction is short-ranged and contains tensor and three-body components that produce dense Pauli representations in conventional encodings. Resource estimates for ab initio nuclear simulation using second-quantized Hamiltonians therefore sit at the edge of current hardware feasibility.

The GeoVac framework [23, 25] exploits angular-momentum selection rules to produce structurally sparse qubit Hamiltonians for molecular electronic structure, with Pauli counts that are orders of magnitude below Gaussian-basis baselines for the same system size. The angular selection rules, which are derived from Wigner $3j$ symbols and the Gaunt integral, are *independent of the radial potential*: they depend only on the single-particle angular basis, not on whether the interaction is Coulombic, Yukawa, or nucleon–nucleon. This paper investigates whether that sparsity carries over to nuclear shell model Hamiltonians.

We report three results. First, the harmonic oscillator shell closures and the Mayer–Jensen magic numbers are reproduced on the GeoVac $(n_r, \ell, m_\ell, \sigma)$ graph via an explicit spin–orbit term. Second, the deuteron and He-4 qubit Hamiltonians, built from the Minnesota NN potential using Moshinsky–Talmi brackets, exhibit Pauli counts that grow sub-linearly in the Hilbert-space dimension — the deuteron has 16 qubits and 688 Pauli terms, while He-4 has the same 16 qubits but only 828 Pauli terms despite a $12.25\times$ larger Hilbert space. Third, a composed nuclear–electronic encoding for deuterium demonstrates that the block architecture used for core–valence molecular systems extends to nucleon–electron systems, though the $10^{13}$ coefficient ratio between nuclear-scale and hyperfine terms makes a single-pass quantum simulation impractical without block partitioning.

We also state the Fock projection rigidity theorem, which identifies the parts of the GeoVac framework that transfer to non-Coulomb systems (the angular machinery) and the parts that do not (the $S^3$ conformal projection). This delineation is the main conceptual contribution of the paper.

The paper is organized as follows. Section II constructs the nuclear shell model on the GeoVac lattice. Section III states the Fock projection rigidity theorem. Sections IV and V present the deuteron and He-4 qubit Hamiltonians. Section VI presents the composed nuclear–electronic encoding. Sections VIII and IX conclude.

## II. NUCLEAR SHELL MODEL ON THE HYPERSPHERICAL LATTICE

### A. Harmonic oscillator basis and shell closures

The standard starting point for the nuclear shell model is the three-dimensional isotropic harmonic oscillator with single-particle energies

$$E_N^{\mathrm{HO}} = \hbar\omega\left(N + \tfrac{3}{2}\right), \qquad N = 2n_r + \ell, \tag{1}$$

and degeneracy $(N+1)(N+2)$ per major shell including spin. Summing these degeneracies yields the HO shell closures

$$2,\ 8,\ 20,\ 40,\ 70,\ 112,\ \ldots$$

These are the correct magic numbers of a pure 3D isotropic oscillator but disagree with the experimentally observed nuclear magic numbers above $N = 20$.

On the GeoVac lattice, each state $(n_r, \ell, m_\ell, \sigma)$ is a graph node with diagonal energy given by Eq. (1). Because the HO shell spacing is uniform in $N$, the HO magic numbers are recovered by simple state counting at any choice of kinetic coupling; they do not depend on the graph off-diagonal structure. This is a numerical confirmation of the packing counting rule from Paper 0 [20] as applied to the HO shell degeneracies.

### B. Spin–orbit term and the real magic numbers

Mayer and Jensen [1, 2] recognized that the observed nuclear magic numbers $28, 50, 82, 126$ require an additional spin–orbit interaction whose sign is *opposite* to the atomic case: the nuclear spin–orbit coupling lowers the $j = \ell + 1/2$ sub-shell below $j = \ell - 1/2$. Adding the Nilsson $\ell(\ell+1)$ term, which mimics the flattening of the Woods–Saxon mean field relative to the pure HO, one writes

$$E(N,\ell,j) = \hbar\omega\left(N + \tfrac{3}{2}\right) - v_{ls}\,\langle \boldsymbol{\ell}\cdot\mathbf{s}\rangle - d_{\ell\ell}\,\ell(\ell+1), \tag{2}$$

with $\langle \boldsymbol{\ell}\cdot\mathbf{s}\rangle = \ell/2$ for $j = \ell + 1/2$ and $-(\ell+1)/2$ for $j = \ell - 1/2$.

A two-parameter scan over $(v_{ls}, d_{\ell\ell})$ identifies the range of parameters for which all seven physical magic numbers $2, 8, 20, 28, 50, 82, 126$ are recovered simultaneously as gaps in the single-particle spectrum. With seven HO shells and the midpoint of the valid region we obtain

$$\frac{v_{ls}}{\hbar\omega} \approx 0.171, \qquad \frac{d_{\ell\ell}}{\hbar\omega} \approx 0.021, \tag{3}$$

which is consistent with the Mayer–Jensen parameters tabulated in standard shell-model references [3, 4]. These are nuclear-physics inputs; the GeoVac framework contributes nothing to their determination. The recovered gaps are not uniform in magnitude: within the physically meaningful region (cumulative occupancy $\leq$ 126; boundaries above this sit at the $n_{\max} = 7$ truncation edge, where the missing $N = 7$ intruder levels distort the spectrum), the six lower magic numbers $2, 8, 20, 28, 50, 82$ are the six *largest* gaps, while 126 appears as a clear but non-dominant gap ($\approx 0.11\,\hbar\omega$), tied with the known sub-shell boundary at 40 and below several intermediate sub-shell gaps $(14, 56, 104, 118)$ — the expected granularity of the two-parameter Mayer–Jensen form at this $n_{\max}$ (gap structure pinned by `tests/test_paper23_magic_gaps.py`). Table I gives the level ordering, with shell closures marked.

TABLE I. Level ordering from Eq. (2) at $v_{ls}/\hbar\omega = 0.171$, $d_{\ell\ell}/\hbar\omega = 0.021$. Shell closures at the experimentally observed magic numbers are marked with $\star$.

| Shell | Levels (in order) | Cumul. |
|---|---|---|
| 1 | $0s_{1/2}$ | $2^\star$ |
| 2 | $0p_{3/2},\ 0p_{1/2}$ | $8^\star$ |
| 3 | $0d_{5/2},\ 1s_{1/2},\ 0d_{3/2}$ | $20^\star$ |
| 4 | $0f_{7/2}$ | $28^\star$ |
| 5 | $1p_{3/2},\ 0f_{5/2},\ 1p_{1/2},\ 0g_{9/2}$ | $50^\star$ |
| 6 | $1d_{5/2},\ 0h_{11/2},\ 2s_{1/2},\ 0g_{7/2},\ 1d_{3/2}$ | $82^\star$ |
| 7 | $1f_{7/2},\ 0i_{13/2},\ 2p_{3/2},\ 0h_{9/2},\ 1f_{5/2},\ 2p_{1/2}$ | $126^\star$ |

The spin–orbit operator is implemented directly in the $(m_\ell, m_s)$ basis via

$$\boldsymbol{\ell}\cdot\mathbf{s} = \ell_z s_z + \tfrac{1}{2}(\ell_+ s_- + \ell_- s_+),$$

producing an off-diagonal coupling between $(m_\ell, \uparrow)$ and $(m_\ell + 1, \downarrow)$ within each $(n_r, \ell)$ subspace. The two-body electron-repulsion-like integrals are unchanged because spin–orbit is a one-body operator. The net effect on single-particle Hamiltonian sparsity is a modest increase from pure diagonal to a block-tridiagonal structure within each $(n_r, \ell)$ submatrix; the overall qubit encoding cost is dominated by the two-body terms in all cases we consider.

## III. THE FOCK PROJECTION RIGIDITY THEOREM

The GeoVac framework for electronic structure uses a conformal projection, originally due to Fock [15], that maps the bound-state spectrum of the hydrogen atom onto the free Laplacian on the three-sphere $S^3$. The projection is

$$p_0 = \sqrt{-2E_n} = Z/n,$$

and it works because the hydrogen spectrum is $\ell$-degenerate within each principal quantum number $n$: all orbitals $(n, \ell, m)$ with the same $n$ have the same energy. The conformal projection then identifies the radial Schrödinger equation on $\mathbb{R}^3$ with the Laplace equation on

$S^3$, and the graph Laplacian on the $(n, \ell, m)$ lattice converges to the continuous $S^3$ Laplacian in the asymptotic limit (Paper 7 [22]).

For nuclear shell model Hamiltonians the projection fails, for reasons that are elementary but worth stating precisely. We summarize the result as

**Theorem (Fock projection rigidity).** *The $S^3$ Fock projection $p_0 = \sqrt{-2E_n}$ maps a one-particle central-field Hamiltonian onto the free Laplacian on $S^3$ if and only if the spectrum $E_{n\ell}$ is independent of $\ell$ within each $n$. The Coulomb potential $-Z/r$ is the unique central potential exhibiting this accidental degeneracy, as a consequence of its hidden SO(4) symmetry (the Runge–Lenz vector). Any deformation $V(r) = -Z/r + \delta V(r)$ with $\delta V \not\equiv 0$ lifts the $\ell$-degeneracy, breaking the conformal equivalence between the radial Schrödinger problem and the $S^3$ graph Laplacian.*

The proof is standard and amounts to observing that Fock's stereographic construction in momentum space requires the hydrogenic relation $E_n \propto -1/n^2$ with no $\ell$-dependence; this fails for any potential that does not have the SO(4) symmetry of the pure Coulomb problem [16]. The consequence for the GeoVac lattice is that the graph node weights, which depend only on $n$ in the Coulomb case, cannot absorb an $\ell$-dependent correction without reducing the graph Laplacian to the trivial diagonal of the corrected spectrum. In that regime the off-diagonal adjacency structure contributes nothing.

The relevant central potentials for nuclear physics — the 3D harmonic oscillator, the Woods–Saxon well, the square well, and the Yukawa potential — all have $\ell$-dependent spectra. The 3D harmonic oscillator has its own accidental degeneracy (the SU(3) group of the isotropic oscillator groups states with common $N = 2n_r + \ell$), but the degeneracy pattern is different from Coulomb's SO(4), and no stereographic projection onto a compact manifold produces the graph Laplacian form that underlies the GeoVac electronic framework. The Bargmann–Segal transform [17] is the natural analogue for the HO, mapping the $L^2$ Hilbert space to a holomorphic reproducing-kernel space on $\mathbb{C}^3$; whether this can be discretized into a graph whose Laplacian converges to the HO Hamiltonian is an open question.

*Negative result, positive implication.* The rigidity theorem tells us what *cannot* be carried over from Papers 7 and 11–15 to the nuclear setting: the conformal projection, the claim that the $S^3$ graph Laplacian *is* the quantum kinetic operator in some dual description, and the $-1/16$ kinetic constant. What *can* be carried over is the angular machinery: the Wigner $3j$ coupling coefficients, the Gaunt integral selection rules, the block structure of the two-body interaction tensor, and the $O(Q^{2.5})$ Pauli scaling that follows [25]. For nuclear shell models, we use the angular machinery to assemble the qubit Hamiltonian and use explicit radial matrix elements (Slater-like integrals computed from HO wavefunctions) for the radial part of the two-body interaction. The radial part is no longer related to the graph; it is a precomputed numerical table.

## IV. THE DEUTERON: FIRST TWO-SPECIES NUCLEAR QUBIT HAMILTONIAN

### A. System and interaction

The deuteron is a bound state of one proton and one neutron in the ${}^3S_1$ channel, with experimental binding energy 2.2246 MeV and an rms charge radius of $\sim 2.1$ fm. The large radius is a consequence of the small binding; the deuteron wavefunction extends well beyond the range of the nuclear force, and accurate binding-energy calculations require oscillator bases with many major shells. We use the Minnesota potential [5], a phenomenological soft-core central interaction composed of three Gaussian terms:

$$V_{\rm NN}(r) = \sum_{i=1}^{3} V_i\, e^{-\kappa_i r^2} \left[\tfrac{1}{2}(u + (2-u)P^\sigma)\right], \qquad (4)$$

with $(V_R, V_T, V_S) = (200, -178, -91.4)$ MeV and the standard $u$ and range parameters of Ref. [5]. The operator $P^\sigma$ is the spin exchange. The two-body matrix elements in the HO basis are computed by transforming the antisymmetrized two-nucleon wavefunction to relative and center-of-mass coordinates using Moshinsky–Talmi brackets [6, 7].

### B. Two-species qubit encoding

Protons and neutrons are distinct fermion species for the purposes of antisymmetrization: Pauli exclusion applies within each species but not between them. We therefore allocate two separate Jordan–Wigner registers, one of size $Q_p$ for the proton orbitals and one of size $Q_n$ for the neutron orbitals, and combine them by tensor product. At $N_{\rm shells} = 2$ (major shells $N = 0$ and $N = 1$) each species has

$$Q_p = Q_n = \underbrace{2}_{0s_{1/2}} + \underbrace{6}_{0p} = 8,$$

giving a total $Q = 16$ qubits. The single-particle states are labeled in the uncoupled $(n_r, \ell, m_\ell, m_s)$ basis; the coupling to $j$-scheme is performed implicitly via the Minnesota matrix elements.

The one-body Hamiltonian is

$$h_1 = \hbar\omega(2n_r + \ell + \tfrac{3}{2})\,\delta_{ij},$$

diagonal in the HO basis. The two-body Hamiltonian is

$$\hat{V}_{\rm pn} = \sum_{ijkl} V_{ijkl}\, a^\dagger_{p,i} a^\dagger_{n,j} a_{n,l} a_{p,k},$$

TABLE II. Deuteron qubit Hamiltonian at $N_{\rm shells} = 2$, $\hbar\omega = 10$ MeV. "Non-$I$" denotes the non-identity Pauli terms. Hilbert-space dimension is $\binom{Q_p}{1}\binom{Q_n}{1} = 64$.

| Quantity | Value |
|---|---|
| $N_{\rm shells}$ | 2 |
| Qubits $Q = Q_p + Q_n$ | $8 + 8 = 16$ |
| Non-$I$ Pauli terms | 688 |
| $Z$-only terms | 80 |
| $XY$-containing terms | 608 |
| 1-norm (non-$I$) | $\approx 383.7$ MeV |
| Hilbert dim. $(1p + 1n)$ | 64 |

where $V_{ijkl}$ is computed from Eq. (4) via Moshinsky–Talmi brackets. Because protons and neutrons are on different registers, each term factorizes as

$$a^\dagger_{p,i} a_{p,k} \otimes a^\dagger_{n,j} a_{n,l},$$

yielding a tensor-product Pauli string that has support on both registers.

### C. Resource counts

Table II summarizes the qubit Hamiltonian at $N_{\rm shells} = 2$, $\hbar\omega = 10$ MeV. The non-identity Pauli term count is 688, with a 1-norm of $\approx 384$ MeV dominated by the HO diagonal and the Minnesota spin-triplet contribution. Of the 688 terms, 80 are $Z$-only (number-operator products) and the remaining 608 carry at least one $X$ or $Y$, reflecting the off-diagonal angular couplings generated by the Minnesota operator.

### D. Binding energy at $N_{\rm shells} = 2$

Diagonalizing the $(Q_p = 8) \times (Q_n = 8) = 64$-dimensional FCI matrix at $\hbar\omega = 10$ MeV and subtracting the HO zero-point energy gives a bound-state energy that is far from the experimental $-2.2246$ MeV. This is expected: the deuteron's rms radius of $\sim 2.1$ fm is comparable to the HO length at $\hbar\omega = 10$ MeV, so $N_{\rm shells} = 2$ has insufficient radial flexibility to describe the long tail of the wavefunction. Converged Minnesota calculations require $N_{\rm shells} \gtrsim 6$–8. We therefore emphasize that the results in Table II are *encoding-validation* benchmarks, not nuclear-structure predictions. The Pauli count and 1-norm characterize the cost of the qubit Hamiltonian; the FCI validation is performed on the matrix representation (the source of truth), while the Pauli-string export is validated structurally, subject to the off-diagonal sign limitation disclosed in Sec. VI D. Recent deuteron-targeted quantum-resource work [11, 12] provides direct comparison points at comparable qubit count for the encoding cost reported here.

TABLE III. Comparison of deuteron and He-4 qubit Hamiltonians at $N_{\rm shells} = 2$. Both systems use 16 qubits; the difference is the Hilbert-space dimension and the number of particles. Pauli counts refer to non-identity terms.

| System | $Q$ | Pauli | Hilbert dim. |
|---|---|---|---|
| Deuteron $(1p + 1n)$ | 16 | 688 | 64 |
| He-4 ($2p + 2n$, no Coulomb) | 16 | 828 | 784 |
| He-4 ($2p + 2n$, + Coulomb) | 16 | 828 | 784 |

## V. HELIUM-4: ANTISYMMETRIZATION WITHIN SPECIES

We extend the deuteron encoding to He-4, a doubly magic nucleus with two protons and two neutrons, experimental binding energy 28.296 MeV. The architecture is the same as for the deuteron, with two changes:

1. Intra-species antisymmetrization: protons fill a 2-particle Slater determinant on the proton register, neutrons fill a 2-particle Slater determinant on the neutron register. The resulting two-body terms include $pp$ and $nn$ antisymmetrized matrix elements in addition to the $pn$ cross-species terms.

2. Coulomb repulsion between the two protons, added as a conventional two-body Coulomb matrix element in the HO basis (approximately 0.7 MeV for the ground state).

Both changes leave the number of qubits unchanged ($Q = 16$) because adding particles does not add spin orbitals; they only increase the Hilbert-space dimension. At $N_{\rm shells} = 2$ the $pp$ pair occupies a 2-particle subspace of dimension $\binom{8}{2} = 28$, and similarly for $nn$, so the full He-4 Hilbert space has dimension $28 \times 28 = 784$.

Table III compares He-4 to the deuteron. The key observation is that the Pauli count grows by only 20% ($688 \to 828$) even though the Hilbert-space dimension grows by a factor of $784/64 = 12.25$. The angular selection rules responsible for the deuteron sparsity carry over to the antisymmetrized pairs without introducing new classes of terms; only the $pp$ and $nn$ blocks are added, and those blocks share the same angular structure as the $pn$ block. The Coulomb correction is a small additional contribution that does not add any new Pauli terms (it modifies only the existing $pp$ coefficients).

The 1-norm of the He-4 Hamiltonian at $\hbar\omega = 10$ MeV is $\approx 512$ MeV (without Coulomb) and $\approx 507$ MeV (with Coulomb), reflecting the dominance of the HO diagonal piece. The ground-state energy from FCI diagonalization is far from the experimental $-28.296$ MeV at $N_{\rm shells} = 2$, for the same reason as in the deuteron: the HO basis is too small to describe the $\alpha$-particle charge distribution. Converged Minnesota calculations of He-4 require $N_{\rm shells} \gtrsim 6$. As with the deuteron, we emphasize that Table III is an encoding-validation benchmark. The physical result of interest is the sub-linear Pauli scaling, which is a direct consequence of the angular-sparsity theorem of Paper 22 [25] and confirms that the theorem applies to multi-particle systems beyond the single-nucleon case.

## VI. COMPOSED NUCLEAR–ELECTRONIC DEUTERIUM

The composed architecture used for core–valence molecular electronic structure [23, 24] extends naturally to the nuclear–electronic case: the nuclear block is the deuteron Hamiltonian from Sec. IV, the electronic block is a single-electron hydrogen Hamiltonian on the $(n, \ell, m)$ lattice at $n_{\max} = 2$, and the coupling between them is given by two one-body-like operators.

### A. Blocks and couplings

**Nuclear block (Block A).** One proton and one neutron in the HO basis at $N_{\text{shells}} = 2$, 16 qubits, as described in Sec. IV. Energy scale $\sim 10^1$–$10^2$ MeV after conversion to atomic units.

**Electronic block (Block B).** A single electron in the hydrogenic $1s$–$2p$ manifold (Coulomb potential $-1/r$), $Q_e = 10$ qubits (5 spatial orbitals $\times$ 2 spins: $1s$, $2s$, $2p_{-1}$, $2p_0$, $2p_{+1}$). The ground state is the exact hydrogenic $1s$ at $-0.5$ Ha.

**Finite-size coupling.** A one-body operator on the electronic register shifts the $1s$ energy by the finite nuclear-size correction $\Delta E_{1s} = (2/5)Z^4 R_{\text{nuc}}^2/n^3$ [18, 19]. At the proton charge radius $R_{\text{nuc}} = 0.8414$ fm, the shift is $\Delta E \approx 1.01 \times 10^{-10}$ Ha, far below the subtraction precision of float64 arithmetic when combined with the nuclear-scale terms. The shift is reported analytically rather than by numerical subtraction.

**Hyperfine coupling.** The proton spin $\mathbf{I}$ couples to the electron spin $\mathbf{S}$ via $H_{\text{hf}} = A_{\text{hf}}\, \mathbf{I} \cdot \mathbf{S}$, where the hyperfine constant for the hydrogen 21-cm transition is $A_{\text{hf}} = 1420.4$ MHz $\approx 2.16 \times 10^{-7}$ Ha. The singlet–triplet gap is $A_{\text{hf}}$, and the maximum shift of any single ground-state manifold member is $3A_{\text{hf}}/4 \approx 1.62 \times 10^{-7}$ Ha.

The composed Hamiltonian is

$$H = H_{\text{nuc}} \otimes I_e + I_{\text{nuc}} \otimes H_e + V_{\text{fs}} + H_{\text{hf}}, \tag{5}$$

with $V_{\text{fs}}$ the finite-size shift on the electronic register and $H_{\text{hf}}$ the cross-register hyperfine operator.

### B. Resource counts and coefficient hierarchy

The composed Hamiltonian uses $Q = 16 + 10 = 26$ qubits. The Pauli count on the full register is 710 non-identity terms, of which 688 come from the nuclear block, 10 from the electronic block diagonal (one $Z$ per spin-orbital plus the absorbed finite-size shift), and 12 from the cross-register hyperfine operator. The finite-size shift is absorbed into the diagonal electronic terms and does not add new Pauli strings.

TABLE IV. Coefficient hierarchy in the composed nuclear–electronic deuterium Hamiltonian (atomic units).

| Contribution | Scale (Ha) |
|---|---|
| Max Pauli coeff (nuclear) | $\sim 5 \times 10^5$ |
| Electronic binding scale | $\sim 10^{-1}$ |
| Hyperfine coeff (per Pauli) | $\sim 10^{-8}$ |
| Finite-size shift (analytic) | $\sim 10^{-10}$ |
| Pauli coeff ratio max/min | $\sim 2 \times 10^{13}$ |

The coefficient hierarchy spans $\sim 13$ orders of magnitude (Table IV): the nuclear diagonal dominates at $\sim 5 \times 10^5$ Ha (after conversion from MeV to Ha via 1 Ha $\approx 27.2$ eV), the electronic binding is $\sim 0.5$ Ha, the hyperfine coefficient is $\sim 10^{-8}$ Ha per Pauli string (the $A_{\text{hf}}/16$ from the $\mathbf{I} \cdot \mathbf{S}$ decomposition), and the finite-size shift is $\sim 10^{-10}$ Ha but does not appear in the Pauli list because it is below the float64 subtraction floor relative to the nuclear-scale diagonal.

### C. Multi-scale precision problem

The $\sim 10^{13}$ coefficient ratio is the central obstacle to running Eq. (5) as a single-pass quantum simulation on near-term hardware. To resolve a hyperfine shift of $10^{-7}$ Ha against a diagonal that contains $10^5$ Ha entries requires an effective energy precision of $\sim 10^{-12}$ in relative terms, which is well beyond the scope of VQE or QPE at current shot counts. The finite-size shift is not only below VQE precision but below the float64 subtraction floor of the nuclear diagonal, and must be computed analytically as a perturbative shift rather than by direct diagonalization.

The practical consequence is that the composed nuclear–electronic Hamiltonian should be solved in block-partitioned form: the nuclear block is solved first (using the deuteron Hamiltonian in isolation), the electronic block is solved second (at fixed proton charge distribution), and the cross-block couplings are treated perturbatively as classical corrections. This is the quantum-chemistry analogue of the Born–Oppenheimer approximation, and it is appropriate whenever the block energies span orders of magnitude. The composed Hamiltonian in Eq. (5) is most useful as an *architectural* object: it lets one write down the full Hamiltonian on a single qubit register, verify correctness by comparison with the block decomposition, and reason about the cost of adding cross-block couplings, even if the single-pass solution is not practical.

### D. Known limitation

The Pauli encoding of the off-diagonal one-body terms in the deuteron builder of Sec. IV carries a known sign issue: the $a_i^\dagger a_j$ operator for $i \neq j$ is currently encoded using $XY - YX$ with a real coefficient, which yields the Hermitian combination $A \otimes A - C \otimes C$ rather than the expected $A \otimes A + C \otimes C$. This affects only the off-diagonal piece of the single-particle Hamiltonian, which in the HO basis is identically zero for the diagonal kinetic term, so the deuteron FCI spectrum is unaffected. The cross-register hyperfine term in the composed Hamiltonian is built via the matrix-form code path in `nuclear_electronic.py`, which bypasses the Pauli builder and yields the correct spectrum directly. We report this as a known issue; it does not affect the Pauli counts or the 1-norms reported in Tables II–IV, which are read from the term list of the encoded Hamiltonian.

### E. Positioning in the noncommutative-geometry literature

The composed nuclear–electronic Hamiltonian of this section is also an explicit Connes-style spectral-triple-level construction in the sense of the GeoVac spectral-triple framework (Paper 32 [26] §V, §VIII.D). The Hilbert space is the tensor product of two real-space registers for two physically distinct quantum particles (proton + electron) at categorically different focal lengths; the Dirac operator carries cross-register coupling ($\mathbf{I} \cdot \mathbf{S}$ hyperfine); and the construction is audited against the canonical Connes axioms by the surrounding Paper 32 framework. The published noncommutative-geometry / spectral-triple lineage (Connes–Marcolli–Chamseddine SM almost-commutative geometry; Connes–van Suijlekom spectral truncations and their convergence theorems) does not contain a directly comparable real-space multi-particle construction: the SM case couples a Riemannian factor with a *finite-dimensional internal* factor, and the truncation-convergence line covers single triples or one-infinite-plus-one-finite products. The structural observation, an explicit literature comparison, and the empirical validation against the 21 cm hyperfine gap are formalized in a standalone Zenodo memo at `papers/group4_quantum_computing/track_ni_spectral_triple_zenodo.md` with cross-references to Paper 32 §V (sub-sector table) and §VIII.D (frontier-of-field framing).

## VII. CROSS-REGISTER $V_{eN}$ AND OPERATOR-LEVEL MAGNETIZATION-DENSITY INNER-FLUCTUATIONS

The composed nuclear–electronic Hamiltonian of Sec. VI couples two real-space registers but treats both the electron–nucleus spatial interaction and the proton magnetization as classical scalars: the finite-size correction $V_{\rm fs}$ is evaluated at a fixed scalar $R_p$ and absorbed into the electronic diagonal, and no operator on the proton register represents the nuclear coordinate. This section reports the construction of two operator-level upgrades that promote each scalar to a bilinear matrix element on the joint $\mathcal{H}_e \otimes \mathcal{H}_p$ register:

1. a cross-register Coulomb operator $V_{eN}(\hat{\mathbf{r}}_e, \hat{\mathbf{R}}_p)$ at distinct Sturmian focal lengths $(\lambda_e, \lambda_n)$, built on the Roothaan 1951 closed form for the bilinear matrix element of $1/|\mathbf{r} - \mathbf{R}|$ between hydrogenic states [30];
2. a magnetization-density inner-fluctuation $\omega_{\rm magn}$, parameterized by a calibrated radial profile $\rho_M(r; r_Z)$ on the proton register, that reproduces the Eides Zemach correction [31] at the operator level rather than as an external scalar substitution.

The two operators share the same joint-register layout, the same diagonal-density Jordan–Wigner Pauli encoding, and the same multi-$\lambda$ Shibuya–Wulfman radial machinery; they differ only in radial kernel ($-Z/|\mathbf{r}_e - \mathbf{R}_p|$ for $V_{eN}$, the magnetization profile $\rho_M$ for $\omega_{\rm magn}$). They realize, on a finite truncation, the structural picture from Phase B of the multi-focal sprint sequence: $V_{eN}$ and $\omega_{\rm magn}$ are categorically distinct *components* of a single composed inner fluctuation $\omega^{\rm comp} = \omega_{\rm recoil} + \omega_{\rm magn} + \cdots$, structurally analogous to how the Connes–Chamseddine inner fluctuation $\omega = \omega_{\rm gauge} + \omega_{\rm Higgs}$ splits in the Standard Model spectral triple [26, 32].

The empirical content of this section is honest about its scope. $V_{eN}$ reproduces the leading-order Bethe–Salpeter recoil correction to the hydrogen 1s energy at 2.86%, and follow-up symbolic analysis (Sec. VII B) shows that this residual is a basis-truncation artifact of the $1s \times 1s$ Sturmian representation, not a missing physical subleading correction. $\omega_{\rm magn}$ reproduces the Eides Tab. 7.3 leading-order Zemach shift at 0.012% residual without any external scalar substitution: the first radial moment $M_1[\rho_M] = r_Z$ enters automatically through the operator construction. The structural interpretation is that the framework computes the operator action given a fixed $\rho_M$, and the calibration ($r_Z$ value, profile family) remains a Layer-2 input from QCD/scattering data, consistent with the structural-skeleton scope pattern documented in CLAUDE.md §3.5.

### A. Cross-register $V_{eN}$ on the joint Sturmian register

The cross-register two-body matrix element

$$M = \left\langle \chi^{\lambda_e}_{n_e \ell_e m_e}(\mathbf{r}_e)\, \chi^{\lambda_n}_{n_n \ell_n m_n}(\mathbf{R}_p) \,\middle|\, \frac{1}{|\mathbf{r}_e - \mathbf{R}_p|} \,\middle|\, \chi^{\lambda_e}_{n'_e \ell'_e m'_e}(\mathbf{r}_e)\, \chi^{\lambda_n}_{n'_n \ell'_n m'_n}(\mathbf{R} \tag{6}$$

is a six-coordinate integral that closes in elementary functions when the proton coordinate $\mathbf{R}_p$ is promoted from a classical scalar to a quantum operator on a Sturmian register at focal length $\lambda_n$. The closure has three pieces:

1. Multipole expansion of $1/|\mathbf{r}-\mathbf{R}|$ in Legendre polynomials with bilateral Gaunt selection on both the electron and the proton angular labels. The expansion terminates exactly at $L_{\max} = \min(\ell_e+\ell_e', \ell_n+\ell_n')$ — the tighter of the two Gaunt bounds.

2. A bilateral angular factor: a product of two Wigner $3j$ symbols, one on each side, with cross-register total-$m$ conservation $m_e+m_n = m_e'+m_n'$.

3. A double radial integral whose inner $R$-integral at fixed $r$ is the multi-$\lambda$ Shibuya–Wulfman split-region integral with combined decay rate $\alpha_{\text{tot}} = \lambda_a+\lambda_b$, evaluated analytically through incomplete gamma functions, and whose outer $r$-integral is performed by Gauss–Laguerre quadrature or, for the simplest case, by a closed form.

For $(n_e,\ell_e) = (n_e',\ell_e') = (1,0)$ and $(n_n,\ell_n) = (n_n',\ell_n') = (1,0)$, $L=0$, the integral collapses to the Roothaan 1951 closed form

$$\boxed{J_0(\lambda_e,\lambda_n) = \frac{\lambda_e\lambda_n\left(\lambda_e^2+3\lambda_e\lambda_n+\lambda_n^2\right)}{(\lambda_e+\lambda_n)^3}} \tag{7}$$

which reproduces the textbook value 5/8 at $\lambda_e=\lambda_n=1$, is symmetric in $(\lambda_e\leftrightarrow\lambda_n)$, and recovers the point-nucleus limit $J_0\to\lambda_e$ as $\lambda_n\to\infty$ at the rate $O(1/\lambda_n^2)$.

*a. Implementation.* The construction is implemented in `geovac/cross_register_vne.py` ($\sim$ 1330 lines, 56 fast tests) as an additive extension of the single-$\lambda$ Shibuya–Wulfman module of Paper 19. Two new low-level functions in `geovac/shibuya_wulfman.py` carry the multi-$\lambda$ generalization: `_hydrogenic_poly_coeffs_lam(lam, n, l)` returns the Sturmian polynomial coefficients $R^\lambda_{n\ell}(r) = e^{-\lambda r}\sum_k c_k r^k$ at independent exponent $\lambda$, and `_radial_split_integral_lam(lam_a, n_1, l_1, lam_b, n_2, l_2, L, R)` computes the mismatched-exponent split-region radial integral. Bit-identical regression to the legacy single-$\lambda$ functions is verified to relative error $\sim 10^{-12}$ across the full single-$\lambda$ test suite (26/26 tests pass without modification).

*b. Verification.* The four structural properties expected of a cross-register Coulomb operator are verified in `tests/test_cross_register_vne.py`: hermiticity ($V[i,j,k,l] = V[k,l,i,j]^*$), block-diagonal structure under cross-register total-$m$ conservation, monotonic convergence of $J_0(\lambda_e,\lambda_n)$ to $\lambda_e$ as $\lambda_n\to\infty$ (verified at $\lambda_n\in\{10,10^2,10^3,10^4,10^5\}$ with relative error scaling as $2/\lambda_n^2$), and bit-identical reproduction of the bare-Coulomb $V_{eN}$ in the limit $\lambda_n\to\infty$. Multipole termination is verified at $L_{\max}=\ell_e+\ell_e'$ across $\ell_e,\ell_e'\in\{0,1,2\}$, preserving the tight Gaunt bound under exponent mismatch.

### B. Bethe–Salpeter leading order and the basis-truncation clarification

The cross-register $V_{eN}$ at $\lambda_e=1$, $\lambda_n = 2\sqrt{M_p/m_e}\approx 85.7\,\text{bohr}^{-1}$, applied to the hydrogen 1s ground state, produces a recoil-like shift

$$\Delta E^{\text{cross}}_{\text{recoil}} = -Z\big[J_0(\lambda_e,\lambda_n)-\lambda_e\big] = +2.65\times10^{-4}\,\text{Ha}, \tag{8}$$

which agrees with the textbook leading-order Bethe–Salpeter recoil $|E_1|\cdot m_e/m_p = +2.72\times10^{-4}$ Ha at the 2.86% level. The same numerical agreement appears against the leading-order term of the Pachucki–Patkóš–Yerokhin 2018 Foldy–Wouthuysen-reduced two-particle Hamiltonian [33].

The choice $\lambda_n = 2\sqrt{M_p/m_e}$ is structural rather than fitted: the value is determined by requiring the leading $O(1/\lambda_n^2)$ Roothaan term in Eq. (7) to match the reduced-mass leading-order shift,

$$+\frac{2Z\lambda_e}{\lambda_n^2} \stackrel{!}{=} \frac{m_e}{m_p}|E_1| = \frac{1}{2M_p} \quad\Longrightarrow\quad \lambda_n = 2\sqrt{M_p/m_e}.$$

The proton's quantum-mechanical zero-point spread in the reduced-mass coordinates is encoded as the Sturmian focal length on the nucleus register, in direct analogy with the use of $\sqrt{Z/n}$ on the electron register.

The 2.86% residual deserves a sharper interpretation than "higher Roothaan terms." Symbolic Taylor expansion of Eq. (7) in $\varepsilon = 1/\lambda_n$ at $\lambda_e=1$ yields

$$J_0(1,1/\varepsilon) = 1-2\varepsilon^2+5\varepsilon^3-9\varepsilon^4+14\varepsilon^5-20\varepsilon^6+27\varepsilon^7-35\varepsilon^8+\cdots \tag{9}$$

with magnitudes $\{2,5,9,14,20,27,35,\ldots\}$ growing quadratically. At calibrated $\lambda_n = 2\sqrt{M_p/m_e}$, the order-$k$ contribution scales as $1/\lambda_n^k = (m_e/m_p)^{k/2}/2^k$, so *odd* $k$ produces a half-integer power of $m_e/m_p$ while *even* $k$ produces an integer power.

Half-integer powers of $m_e/m_p$ are structurally absent from the Pachucki–Patkóš–Yerokhin Foldy–Wouthuysen expansion of the two-particle Hamiltonian: their series is a polynomial in the single small parameter $m_e/m_p$ with integer-power-only terms. The Roothaan odd-$k$ tower is therefore not a sub-leading recoil correction at all, but a basis-truncation artifact of the $1s\times1s$ Sturmian representation at fixed $n_{\max}=1$. Numerically, the half-integer tower at $\lambda_n = 85.7$ sums to $-7.95\times10^{-6}$ Ha, exactly $-2.92$% of the Bethe–Salpeter leading order, accounting for nearly all of the 2.86% drift.

The complementary integer-only sub-sum, $\sum_{k=2,4,6,8,\ldots} c_k/\lambda_n^k$, lands at $+2.72475\times10^{-4}$ Ha, a $+0.061$% deviation from Bethe–Salpeter. We flag this sub-percent agreement as *fortuitous*, not structural: the order-$k=4$ Roothaan coefficient is $+9$, while the physical Pachucki next-order coefficient is $-1/(2M_p^2)$ — the magnitudes are comparable but the *signs are opposite*. The integer-only filter accidentally lands close because $1/\lambda_n^4$ is small ($\sim 10^{-7}$ Ha) at the calibrated focal length, not because the coefficients are correct. At larger $Z$ where the next-order recoil grows as $Z^5$, the fortuitous cancellation degrades.

The structural conclusion is sharper than the numerical one: the cross-register $V_{eN}$ at $n_{\max} = 1$ matches Pachucki 2018 *leading-order verbatim*, with sub-leading content *basis-truncation-bounded*; closing the next-order sign disagreement structurally requires multi-shell expansion to $n_{\max} \geq 2$ on both registers, which is named as the natural follow-up but is out of scope for this paper. The single-parameter calibration $\lambda_n = 2\sqrt{M_p/m_e}$ is structurally a leading-order-only fix; matching higher orders requires more basis functions, which is consistent with the Layer-2 calibration interpretation of focal length in Paper 18 §IV [27].

### C. Operator-level magnetization-density inner-fluctuation $\omega_{\rm magn}$

The Eides Zemach correction to the hydrogen hyperfine splitting [31] is conventionally written as a scalar substitution

$$\frac{\Delta\nu_Z}{\nu_F} = -2Z\alpha m_e\, r_Z, \qquad r_Z = \int d^3r\, d^3r'\, \rho_E(r')\, \rho_M(|\mathbf{r}-\mathbf{r}'|), \tag{10}$$

where $\rho_E$ and $\rho_M$ are the unit-normalized proton charge and magnetization densities and $r_Z$ is the first moment of their convolution. Sprint HF-4 (Sec. VI, footnote; Sprint HF [38] for the original 21 cm precision test) closed the framework's residual to the 21 cm transition by treating $r_Z$ as a Layer-2 external input scalar; that closure landed at +18 ppm, consistent with the Eides Tab. 7.3 budget for multi-loop QED plus nuclear polarizability.

The construction reported here promotes the scalar substitution to an operator-level inner-fluctuation on the joint $\mathcal{H}_e \otimes \mathcal{H}_p$ register. The operator

$$\omega_{\rm magn} = -2Zm_e \int d^3R\, \rho_M(R)\, \hat{r}_Z[R] \tag{11}$$

is built from a unit-normalized radial profile $\rho_M(r; r_Z)$ that is itself parameterized by the Zemach radius as a Layer-2 calibration scalar. The bilinear matrix element on the joint register at the leading multipole $L = 0$ collapses to

$$\langle \hat{r}_Z \rangle = \int d^3R\, \rho_M(R)\, R + O(r_Z/a_0)^2 = M_1[\rho_M] + O(r_Z/a_0)^2 = r_Z + O(r_Z/a_0)^2 \tag{12}$$

The Eides scalar substitution emerges automatically: the first radial moment of the magnetization density *is* the Zemach radius by the Eides definition, and the operator-level construction recovers it through the standard hydrogenic 1s contact integral, with no external substitution required. Higher-order corrections in $r_Z/a_0$ are the Friar moment $\langle r^2 \rangle_{\rho_M}$ and beyond, suppressed by $\sim 7.8 \times 10^{-6}$ relative to the leading shift on hydrogen.

*a. Implementation.* The module `geovac/magnetization_density.py` ($\sim$ 480 lines, 27 fast tests) supports three radial profiles: Gaussian $\rho_M(r) = (\beta/\pi)^{3/2}\, e^{-\beta r^2}$ with $\beta = 4/(\pi r_Z^2)$, exponential $\rho_M(r) = (\kappa^3/8\pi)\, e^{-\kappa r}$ with $\kappa = 3/r_Z$, and a delta limit. All three are unit-normalized ($M_0 = \int d^3r\, \rho_M = 1$) and calibrated to the same first moment ($M_1 = r_Z$ for Gaussian and exponential by construction). They differ only at $M_2$, the Friar moment, which controls the sub-leading shape-dependent correction.

*b. Verification.* At the Eides calibration $r_Z = 1.045\,\text{fm} = 1.974 \times 10^{-5}$ bohr, the operator-level construction gives

$$\left.\frac{\Delta\nu_Z}{\nu_F}\right|_{\rm operator} = -39.495\,\text{ppm}, \tag{13}$$

versus the Eides Tab. 7.3 reference $-39.500$ ppm, a residual of +0.005 ppm or 0.012% of the Eides shift. The residual is consistent with the rounding of the bohr-to-fermi conversion constant $a_0 = 52917.72108$ fm in the input scalar; with a sharper conversion the agreement lands at machine precision. Profile independence at leading order is verified to machine precision: Gaussian and exponential profiles, both calibrated to the same $r_Z$, produce identical leading-order shifts. They differ at order 2 by $\sim 13\%$ (Gaussian $M_2 = 4.59 \times 10^{-10}$ bohr$^2$ vs. exponential $M_2 = 5.20 \times 10^{-10}$ bohr$^2$), well below any precision target relevant to hydrogen 21 cm.

*c. Composition with $V_{eN}$.* On a minimal joint register ($1s_e \times 1s_p$, $Q = 2$ qubits), both $V_{eN}$ and $\omega_{\rm magn}$ produce four-string Pauli sums with shared support on $\{II, ZI, IZ, ZZ\}$. The combined Hamiltonian $H_{\rm combined} = V_{eN} + \omega_{\rm magn}$ is hermitian by linearity, and the Pauli sum is additive. The $\omega_{\rm magn}$ contribution is suppressed relative to $V_{eN}$ by the dimensionless ratio $r_Z/a_0 \sim 2 \times 10^{-5}$. This realizes Phase B of the multifocal sprint sequence's structural expectation that $\omega_{\rm magn}$ is W1a's inner-fluctuation sibling: it shares the joint-register layout, the diagonal-density JW encoding, and the multi-$\lambda$ Shibuya–Wulfman infrastructure with $V_{eN}$, differing only in the radial kernel ($-Z/|\mathbf{r}_e - \mathbf{R}_p|$ vs. $\rho_M$).

The structural reading is that $\omega^{\rm comp} = \omega_{\rm recoil} + \omega_{\rm magn} + \cdots$ is the multi-focal-composition analog of the Connes–Chamseddine inner fluctuation $\omega = \omega_{\rm gauge} + \omega_{\rm Higgs}$ on the SM almost-commutative spectral triple [26, 32]: each component is built from a fixed parameterization of the proton's internal structure ($\rho_E$ for $\omega_{\rm recoil}$, $\rho_M$ for $\omega_{\rm magn}$), the framework computes the operator action on the joint Hilbert space, and the choice of profile family and calibration scalars is Layer-2 input from QCD/scattering data, in the sense of Paper 18 §IV's inner-factor input data tier [27].

### D. Test on deuterium 1S hyperfine structure

The composed nuclear–electronic architecture of this section is designed for $I = 1/2$ proton hyperfine structure (the 21 cm hydrogen test case of Sec. VI). A natural extension is to $I = 1$ deuteron HFS: the cross-register $V_{eN}$ machinery is unchanged (Roothaan 1951 multipole expansion is angular, independent of nuclear spin), the magnetization-density operator $\omega_{\rm magn}$ is parameterized by a scalar moment $r_Z(D)$, and only the hyperfine-coupling Hamiltonian $A_{\rm hf}\,\hat{\mathbf{I}}\cdot\hat{\mathbf{S}}$ acquires a different Clebsch–Gordan multiplicity factor $(2I+1)/2 = 3/2$ between the $F = I + 1/2$ and $F = I - 1/2$ levels.

The framework-native Bohr–Fermi prediction at strict point-nucleus, no-recoil order is

$$\nu_{\rm HFS}^{\rm BF,strict}(D, 1S) = \frac{3}{2} \times \frac{4}{3}\,\frac{g_d^{\rm atomic}}{2}\,\alpha^2 \left(\frac{m_e}{m_p}\right) \text{Ha}, \quad (14)$$

where $g_d^{\rm atomic} = 2\mu_d/\mu_N = 1.7148$ is the atomic-physics Hamiltonian-convention $g$-factor (twice the CODATA value $g_d = \mu_d/(\mu_N \cdot I) = 0.857$ because of the factor $2I$); the divisor restoring $\mu_I/(I\mu_N)$ is 2, not $2I$ — the two coincide only at $I = 1$. The mass factor is $m_e/m_p$ for every nucleus, because the nuclear magneton $\mu_N = e\hbar/(2m_p)$ is defined with the proton mass. With CODATA constants this yields 327.2350 MHz vs the Wineland–Ramsey 1972 [35] measurement of 327.384352522(2) MHz, a residual of −456 ppm. (Corrected 2026-08-28: this equation previously carried $m_e/m_d$ in place of $m_e/m_p$ with no factor-of-2 divisor. Since $m_d/m_p = 1.99900750 \approx 2$ the two errors nearly cancelled, leaving the baseline high by 496.5 ppm; $I = 1/2$ nuclei were unaffected, so the H and T cross-checks did not expose it.)

The leading-order Zemach correction with $r_Z(D) = 2.593$ fm (Friar–Payne 1997 [36]) gives $\Delta\nu_Z/\nu_F = -98.0$ ppm at the operator level via `geovac.magnetization_density.hydrogen_zemach_eides_leading_order`, matching the analytic Eides $-2Zm_e r_Z$ to floating-point precision. Profile (Gaussian vs. exponential) independence is preserved at leading order, identical to the hydrogen regression of Sec. VII C. The cumulative chain (BF + recoil + Schwinger $a_e$ + leading Zemach) lands at 327.3153 MHz with −211 ppm residual against experiment. The chain sits below the Pachucki–Yerokhin / Karshenboim total theory (327.339 MHz), the expected direction for a chain omitting the deuteron polarizability (+44 ppm, Pachucki 2011 [37], QCD-internal NN dynamics) plus multi-loop QED (LS-8a wall, ∼few ppm) plus recoil NLO and finite-size charge corrections.

*a. Multi-focal architecture verdict at $I = 1$.* The Bohr–Fermi residual of −456 ppm is the cleanest framework-native verification of the $I = 1$ cross-register angular coupling structure in the catalogue. The Hamiltonian $A_{\rm hf}\,\hat{\mathbf{I}} \cdot \hat{\mathbf{S}}$ is unchanged between $I = 1/2$ and $I = 1$; only the Clebsch–Gordan multiplicity differs. The deuteron's nonzero electric quadrupole moment $Q_d = 0.286$ efm$^2$ couples to electronic charge-density gradients but enters HFS at sub-ppm level for s-states (no orbital angular momentum to couple to) and is negligible at the precision tested here. At leading order the multi-focal architecture handles $I = 1$ identically to $I = 1/2$, and the only sub-leading mechanism that genuinely opens up at $I = 1$ (the deuteron quadrupole) sits below the Layer-2 input threshold for s-state observables.

Sprint provenance: `debug/precision_catalogue_deuterium_hfs`, `debug/precision_catalogue_deuterium_hfs_memo.md`, `debug/data/precision_catalogue_deuterium_hfs.json`. Operator-level four-component Roothaan autopsy (May 9 2026): `debug/d_hfs_autopsy_track5_memo.md`, which elevates the $I = 1$ verdict from energy level to operator level via explicit construction of $\hat{I} \cdot \hat{S}$ on the 6-dim joint nuclear-electronic Hilbert space (multiplicity ratio $D/H = 3/2$ at machine precision) and reproduction of the Eides analytic leading-order Zemach scalar at $\sim 1.45 \times 10^{-14}$% of the LO shift via the production magnetization-density operator at $r_Z(D) = 2.593$ fm. Cross-references: Paper 34 §V (catalogue row for $\nu_{\rm HFS}(D)$ at −456 ppm BF strict and −211 ppm full chain), Paper 34 §V.C.X (`sec:autopsy_d_hfs` operator-level autopsy), Sprint MH Track B (electronic→muonic mass swap at unchanged proton; this sprint is the spin-1/2→spin-1 nucleus swap at unchanged electron), Sprint HF Track 4 (hydrogen $r_Z$ Layer-2 input, 0.012% Eides regression).

### E. Synthesis and forward references

The cross-register $V_{eN}$ and $\omega_{\rm magn}$ operators together close the operator-level scaffolding of two of the multi-focal walls (W1a and W1b) named in the May 2026 sprint sequence. The honest verdicts are scope-bounded:

- $V_{eN}$: leading-order Bethe–Salpeter recoil verbatim (calibration-fixed); sub-leading content basis-truncation-bounded at $n_{\rm max} = 1$, with the 2.86% drift identified as the half-integer-mass-ratio artifact tower (structurally absent in Pachucki 2018). Multi-shell $n_{\rm max} \geq 2$ on both registers is the structural follow-up for higher-order recoil.

- $\omega_{\rm magn}$: Eides Tab. 7.3 leading-order verbatim at 0.012% residual, with profile independence at leading order and shape dependence appearing only at the Friar moment $\langle r^2\rangle_{\rho_M}$. Higher-order Eides corrections (polarizability $\Delta_{\rm pol}$, recoil-corrected Zemach, Friar moment $\langle r^3\rangle_{(2)}$) each enter as additional inner-fluctuation components in the same architectural slot.

*a. Cross-references.* The convergence theorem under the single-factor Camporesi–Higuchi truncation that supplies the algebraic foundation under the electron register is established in Paper 38 [28], the van Suijlekom state-space Gromov–Hausdorff convergence for the truncated $S^3$ spectral triple. The tensor-product extension that supplies the algebraic foundation under the joint $\mathcal{H}_e \otimes \mathcal{H}_p$ register — the two-infinite-factor convergence theorem that surrounds the cross-register operator construction reported here — is the subject of Paper 39 (in preparation, parallel sprint) [29]. The standalone NCG-positioning observation for the Track NI base construction, including the structural literature gap against the published Connes–Marcolli–Chamseddine and Connes–van Suijlekom lineages, is the Track NI Zenodo memo [34].

*b. Open follow-ups.* Three downstream sprints are named but not pursued in this paper: (i) multi-shell $n_{\max} \geq 2$ on both joint registers, to close the next-order sign disagreement against Pachucki 2018 [33] and lift the basis-truncation ceiling on $V_{eN}$; (ii) full Pachucki $(Z\alpha)^4$ match in the multi-shell extension, requiring the integer-power coefficients to agree term-by-term rather than only at leading order; (iii) $n_e$-projected FCI on the joint register including the Slater–Condon decomposition needed for many-electron antisymmetrization, which is the structural extension of the diagonal-density encoding used here. Each is architecturally compatible with the present construction; each is a scoped follow-up sprint.

## VIII. DISCUSSION

The results of Sections II–VI suggest a natural partition of the GeoVac framework into a universal component and a Coulomb-specific component:

- **Universal (applies to HO, Woods–Saxon, NN potentials).** The angular selection rules from Wigner $3j$ symbols and the Gaunt integral; the angular-sparsity reduction of single-center Pauli scaling below the naive $O(Q^4)$ (measured $O(Q^{3.77})$ for Coulomb atoms — a narrow margin under the exact global-$M_L$ rule) and the exactly linear composed Pauli scaling in $Q$ at fixed basis; the block structure of the two-body tensor; the qubit encoding via Jordan–Wigner on species-specific registers.
- **Coulomb-specific.** The $S^3$ conformal projection $p_0 = \sqrt{-2E_n}$; the SO(4) hidden symmetry and the Runge–Lenz vector; the $-1/16$ kinetic constant; the Hopf bundle structure underlying the fine-structure constant observation [21].

The nuclear shell model fits squarely in the universal category. The Fock rigidity theorem makes this precise: the conformal projection simply does not survive the transition from Coulomb to HO or Woods–Saxon, so any claim that the GeoVac $S^3$ machinery extends directly to nuclei would be incorrect. What does extend is the angular machinery — and the results of Sections IV and V demonstrate that this machinery is sufficient to produce structurally sparse qubit Hamiltonians for simple nuclei.

*Directions for follow-up work.* (i) Larger $N_{\text{shells}}$ for converged binding energies; the interesting physics question is whether the sub-linear Pauli scaling persists as the basis grows, or whether the Minnesota potential's short range eventually fills in the angular zeros. (ii) Three-body forces (chiral EFT at N$^2$LO); these introduce new classes of Pauli terms beyond the two-body pattern studied here, and their effect on sparsity is an open question. (iii) Larger nuclei such as $^6$Li, $^{12}$C, and $^{16}$O; the composed architecture should apply, with nucleon blocks labelled by single-particle $(n_r, \ell, j)$ orbitals. (iv) Hardware implementation and comparison with NCSM benchmarks on actual quantum simulators; the 16-qubit deuteron is within reach of current ion-trap and superconducting platforms.

A final note on possible Coulomb-free projections. The Bargmann–Segal transform [17] gives a natural holomorphic representation of the HO Hilbert space, in which the HO Hamiltonian becomes a first-order differential operator. Whether the Bargmann–Segal space admits a discrete graph structure whose Laplacian converges to the HO Hamiltonian — a strict HO analogue of the $S^3$ Fock projection — is an open question that we leave for future work.

## IX. CONCLUSION

We have constructed qubit Hamiltonians for the nuclear shell model on the GeoVac hyperspherical lattice. The HO shell closures and Mayer–Jensen magic numbers are reproduced from the graph spectrum with standard spin–orbit and orbital-correction parameters. The deuteron and He-4 qubit Hamiltonians, built from the Minnesota potential via Moshinsky–Talmi brackets, show sub-linear Pauli scaling: He-4 requires only 20% more Pauli terms than the deuteron despite a 12.25× larger Hilbert space. The composed nuclear–electronic deuterium encoding demonstrates that the multi-block architecture extends to nucleon–electron systems but faces a $\sim 10^{13}$ coefficient ratio that requires block-partitioned simulation. The Fock projection rigidity theorem precisely delineates which parts of the GeoVac framework transfer to non-Coulomb systems (the angular machinery) and which do not (the $S^3$ conformal projection). The binding energies reported here at $N_{\text{shells}} = 2$ are not converged and should not be read as nuclear-structure predictions; they are encoding-validation benchmarks that verify the qubit Hamiltonian reproduces the matrix representation.

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